SHIFT LEFT: ELECTRONICS ARCHITECTURES FOR THE NEW SOFTWARE-DEFINED VEHICLE AND S32K MICROCONTROLLERS

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NOVEMBER 2023

SECURE CONNECTIONS
FOR A SMARTER WORLD

PUBLIC

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OVERVIEW

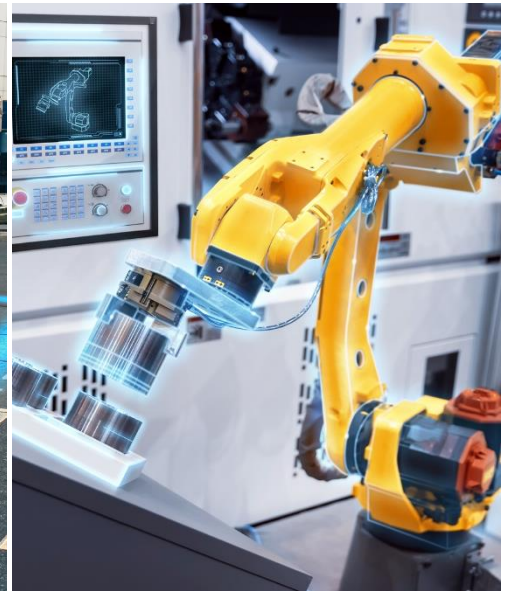
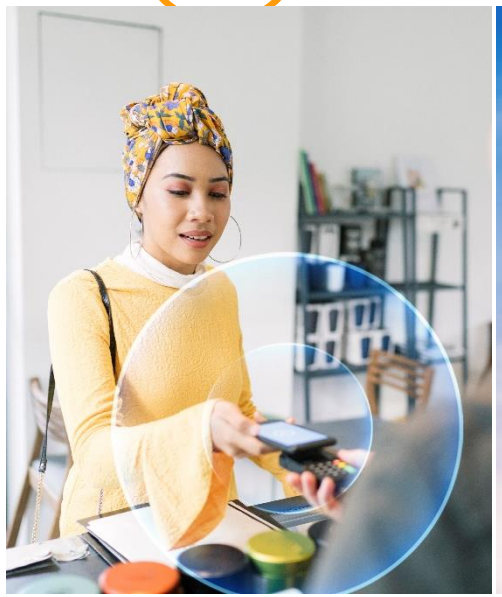
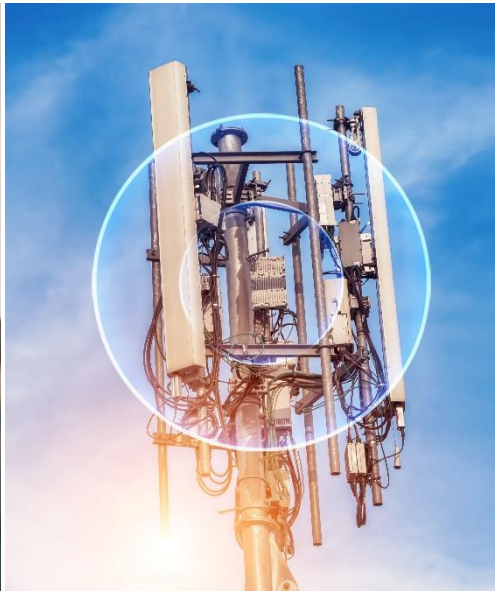
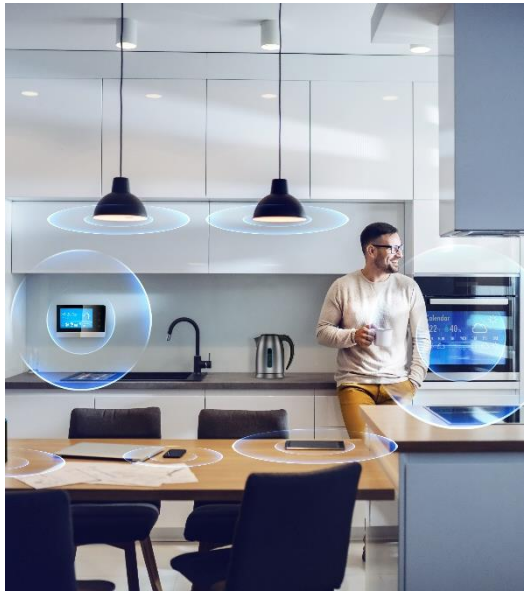
- Software-Defined Vehicle
- Electrical and Electronic Architecture
- Re-Architect Software
- Embrace Virtualization
- Software and Partner
- S32K Microcontrollers
- Q & A

Software-Defined Vehicle

SECURE CONNECTIONS FOR A SMARTER WORLD

OUR DIGITALLY ENHANCED WORLD IS EVOLVING TO ANTICIPATE AND AUTOMATE

NXP Semiconductors N.V. (NASDAQ: NXPI) enables a smarter, safer and more sustainable world through innovation. As the world leader in secure connectivity solutions for embedded applications, NXP is pushing boundaries in the automotive, industrial & IoT, mobile, and communication infrastructure markets.

THREE FUTURE CAR MISSIONS → ONE CLEAR PATH TOWARDS SOFTWARE-DEFINED VEHICLE

CLOUD

Over-the-Air (OTA) Services

Autonomous
New sensing, thinking

Connected & Service Oriented
New E/E architecture

Electric
New energy management

WHAT IS A SOFTWARE-DEFINED VEHICLE?

Software …

defines the customer experience

becomes the differentiator

is the focus point for innovation

brings OEMs new revenue streams

Hardware …

is a stable, scalable future-proof platform

New features include …

car & cloud integration

over-the-air updates

flexibility to continuously evolve

service-oriented architectures

virtualization and containerization

New processes are …

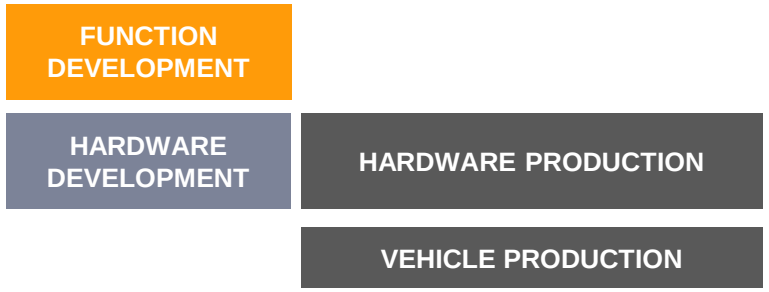
DevOps: continuously integrate, test and deployment of software

Agile and lean methods

digital twin virtual modelling

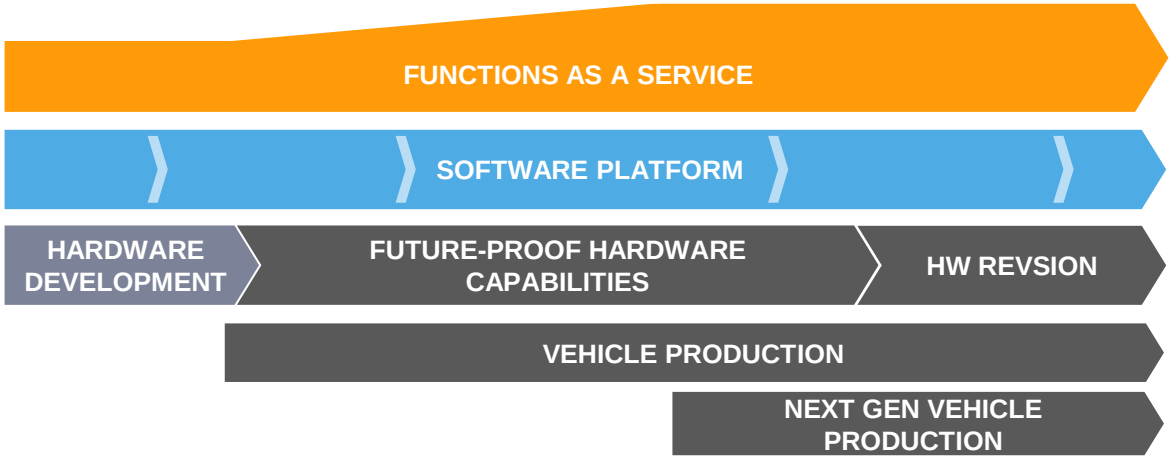
WHAT DOES A SOFTWARE-DEFINED VEHICLE OFFER?

LEGACY
SINGLE HARDWARE AND SOFTWARE DEVELOPMENT



Limits frequency of updates

NEW DECOUPLED HARDWARE AND SOFTWARE DEVELOPMENT



Allows for multiple software updates and code re-use across hardware

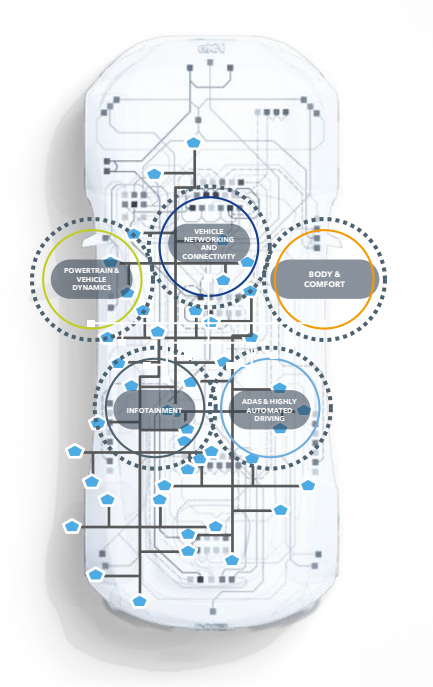
Electrical and Electronic Architecture

E/E ARCHITECTURE EVOLUTION

Flat

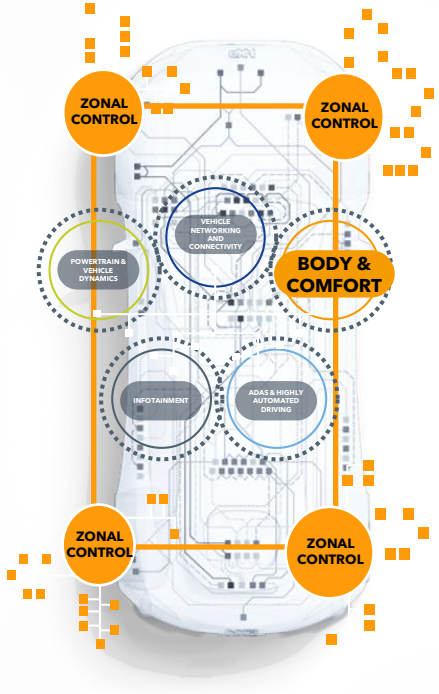
LEGACY ARCHITECTURE ORGANICALLY GROWN

Domain



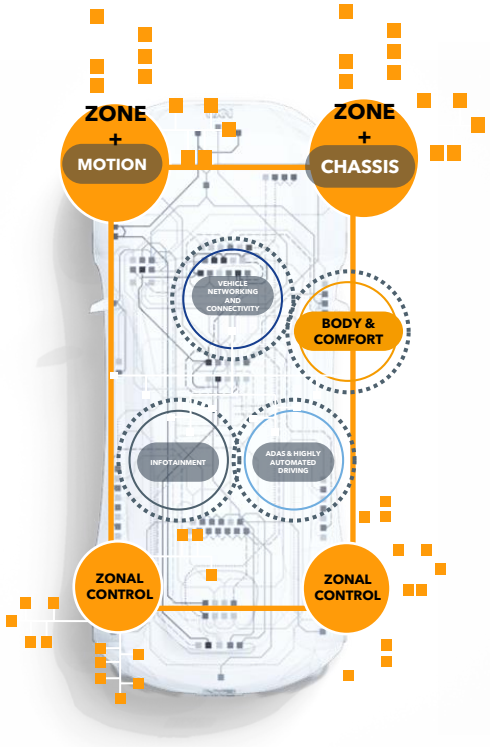
SCALABLE AND CENTRALIZED SOFTWARE DEVELOPMENT

Domain + Body Zonal



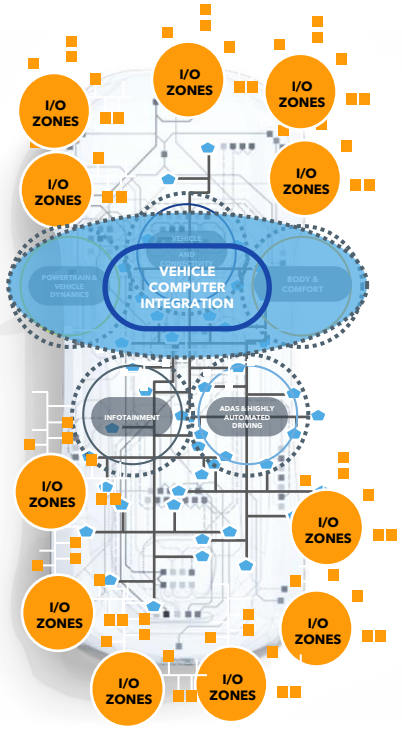
DOMAIN SW BENEFIT + SIMPLIFIED WIRING AND VEHICLE NETWORK

Cross-Domain Zonal



LOWER HW COST VS MORE COMPLEX SW

Consolidated Compute

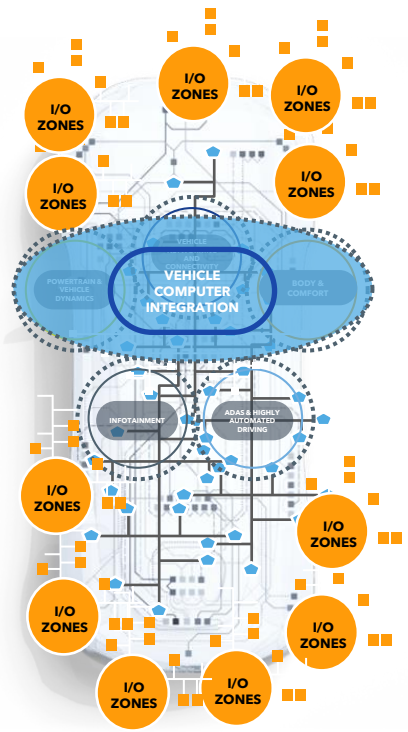


SIGNIFICANTLY REDUCES SW DEVELOPMENT COST, VS HIGHER HW COST FOR FUTURE-PROOFING

LOGICAL AND PHYSICAL TRANSITIONS

Flat

Consolidated Compute



TWO PARALLEL ARCHITECTURAL CHANGES

1 CENTRAL DOMAIN FOCUS: SCALABLE AND CENTRALIZED SOFTWARE DEVELOPMENT

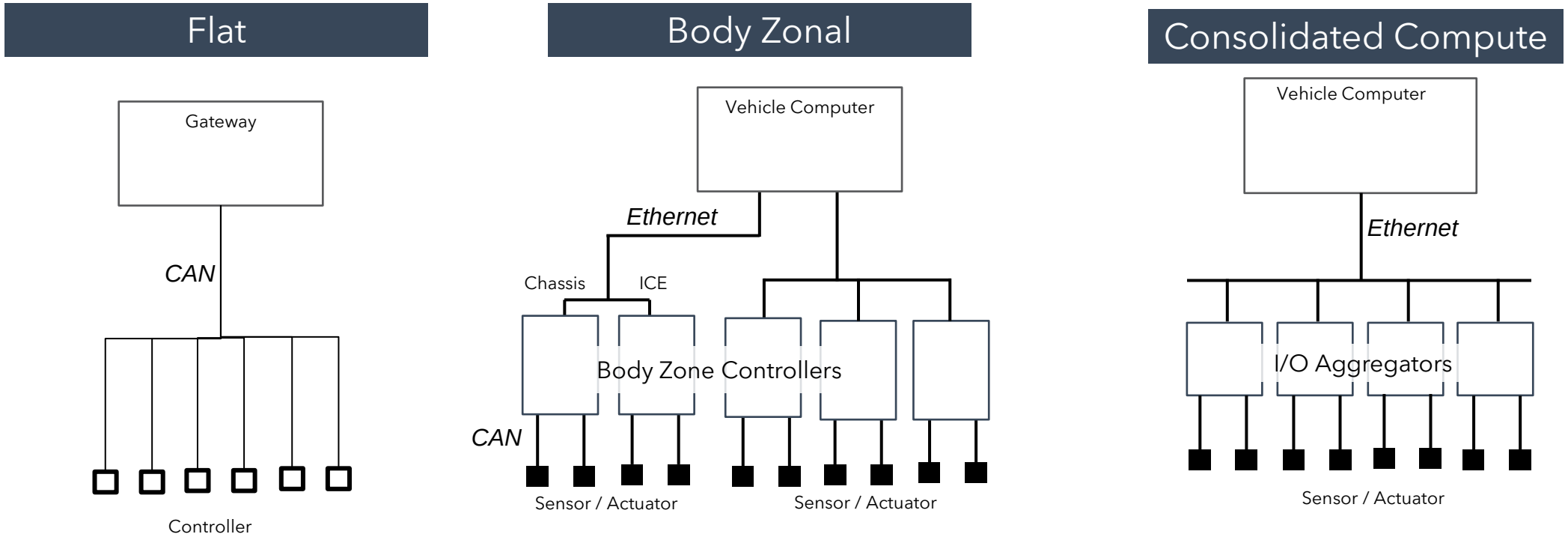
1. Flexible & scalable software environment
2. Efficiently supports the user defined vehicle
3. Centralized OTA, easily upgradable Software

+

2 ZONAL FOCUS: SIMPLIFIED WIRING AND CONNECTIVITY

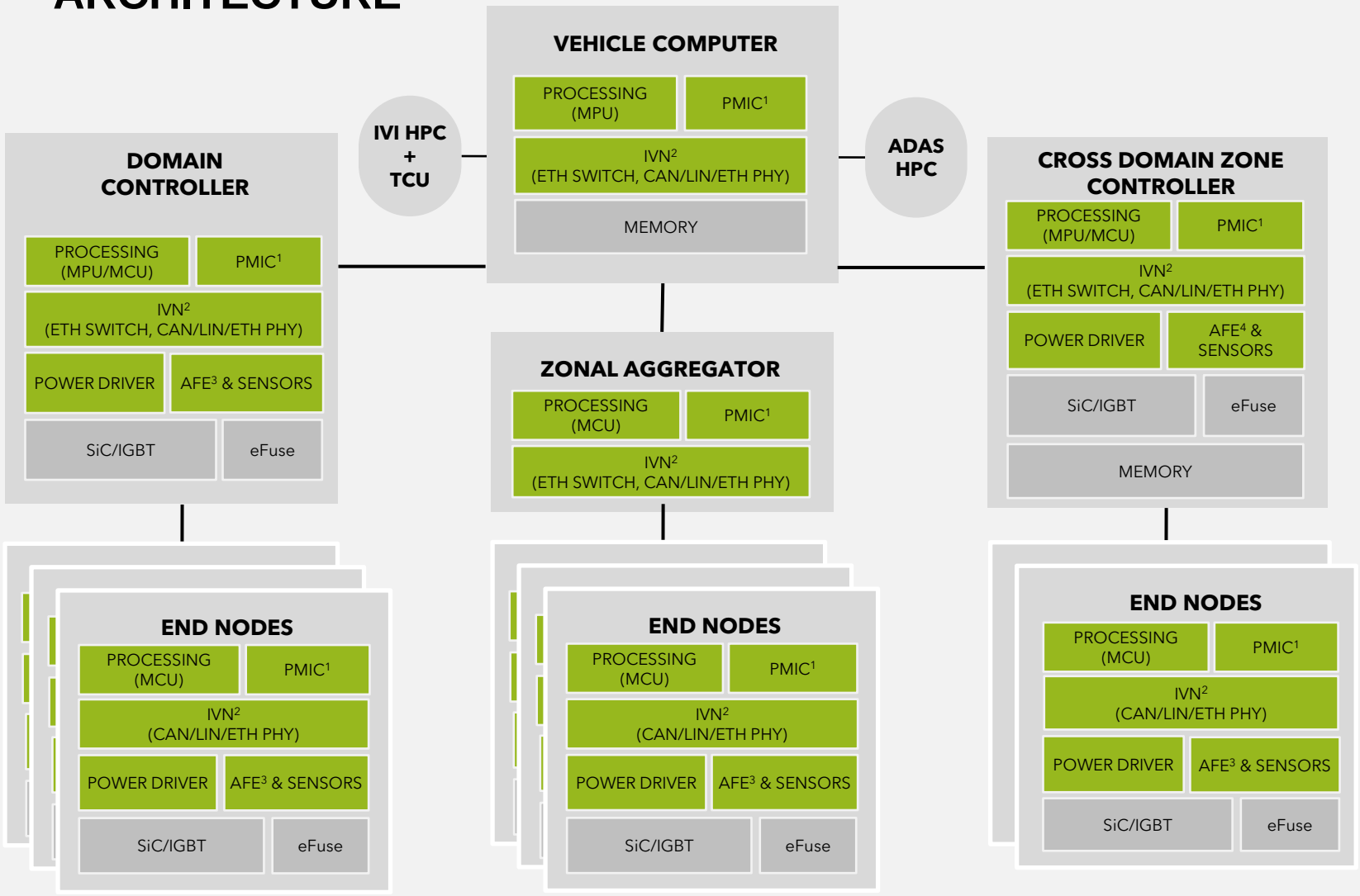
1. Dramatically reduced wire routing and cable costs
2. Flexible data monetization
3. Easily upgradable Hardware

IMPLICATIONS OF E/E ARCHITECTURE CHANGES



- Communications – transition to high bandwidth Ethernet
- Function distribution – dedicated controller at the edge, to hierarchical setup
- ECU consolidation – one vehicle function = one ECU is history

NXP PLATFORM PLAY IN EVOLVING VEHICLE ELECTRONICS ARCHITECTURE



VEHICLE COMPUTER

- **Arm-based safe and secure processors** with industry's highest application and real-time ASIL-D performance, highest security, Multi-Gb ETH networking interfaces and accelerators and AI/ML capabilities
- **PMICs/SBCs** for safety and power management
- Robust, reliable and secure **ETH/CAN/LIN** solutions

ZONAL, DOMAIN, CROSS-DOMAIN MODULES

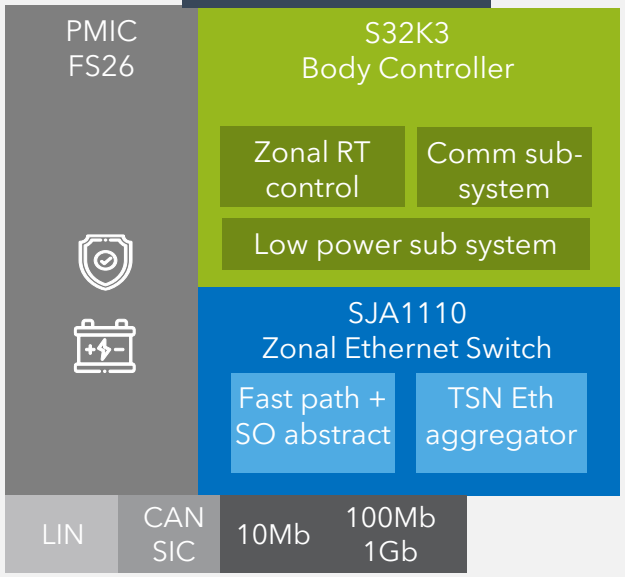
- **Arm-based safe and secure MCUs and processors** with industry's highest ASIL-D performance, highest security, high-end networking interfaces and accelerators and AI/ML capabilities
- **PMICs/SBCs** for safety and power management
- Robust, reliable and secure **ETH/CAN/LIN** solutions
- **Power drivers** for motors, lighting and relays
- **AFE and sensors** for powertrain and motor control

EDGE NODES

- **Arm-based safe and secure MCUs** with industry's highest ASIL-D performance, highest security, low-power, high-end networking interfaces and accelerators
- **PMICs/SBCs** for safety and power management
- Robust, reliable and secure **ETH/CAN/LIN** solutions
- **Power drivers** for motors, lighting and relays
- **AFE and sensors** for powertrain and motor control

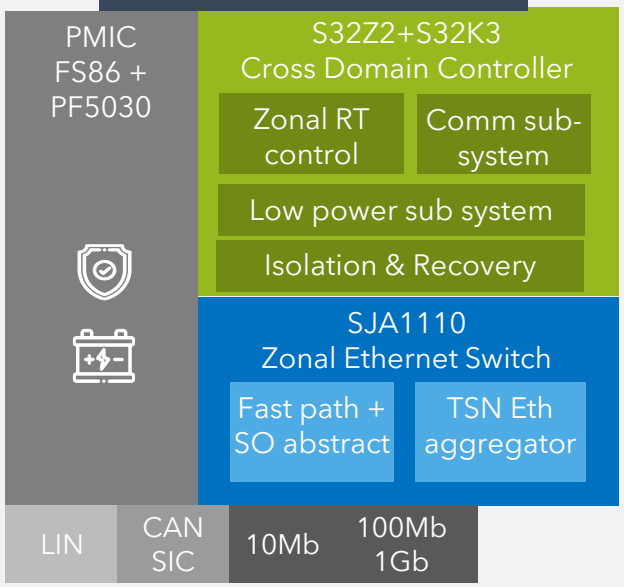
NXP VEA SOLUTIONS LANDSCAPE

Body Zonal



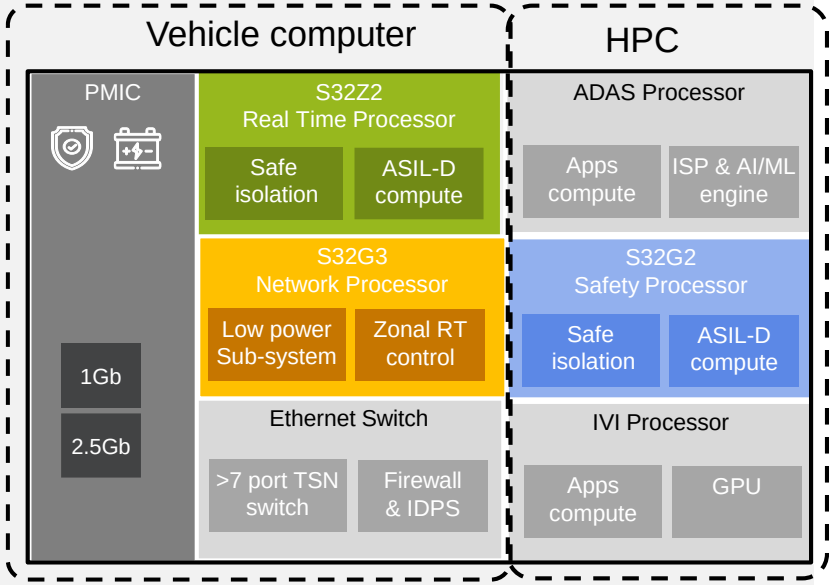
- Three-chip solution (plus PHYs)
- High port and I/O processing scalability
- Smart power distribution, fast wake up
- Strong safety + security

Cross-Domain Zonal



- Three-chip solution (plus PHYs)
- Strong on-chip isolation & recovery
- High port and CAN scalability
- Powerful network, smart power distribution
- Strong safety + security

Consolidated Compute



- Multi-chip full-function HPC solution
- Can be built as single box (connected by PCIe and/or Ethernet) or multiple boxes
- Strong safety + security

COMMON CONFIGURATION & SOFTWARE

Re-Architect Software

RE-ARCHITECTING SOFTWARE

RT COMPUTE

Function

GOAL

Not bound by strict functional or geographical constraints

Maximum flexibility to co-locate these functions, less HW dependent, quicker to update & deploy

First Step:
Identify how to split existing application software into RT compute vs I/O control

I/O CONTROL

Function

GOAL

I/O interface is application agnostic

Multi-consumers of sensors via an API call or service subscription

(e.g. Video stream, temp sensor, …)

I/O Interfaces

Networked (CAN, Eth) ← Network — Direct → **Raw I/O** (ADC, Timer, Pixel)

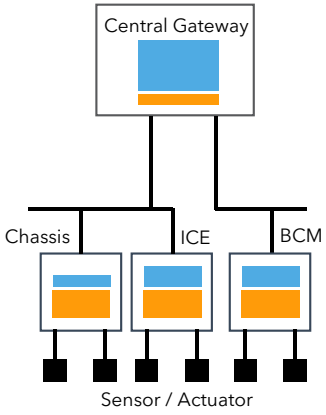
Control Loops

Loosely Coupled → Higher — Lower → **Tightly Coupled**

SW PARTITIONING – ARCHITECTURAL CHOICES

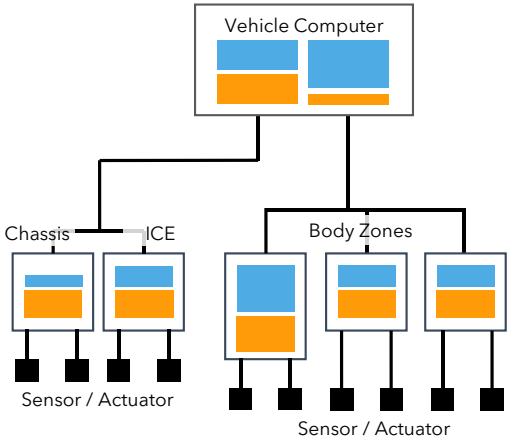
Real Time Compute | I/O Control

Flat



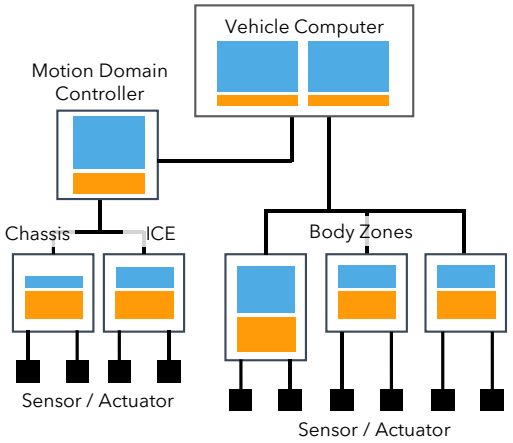
Legacy ECUs
Features enabled with new hardware

Body Zonal



Body Zonalization
- Wiring harness reduction, modularity introduced
- Primary / secondary body controllers

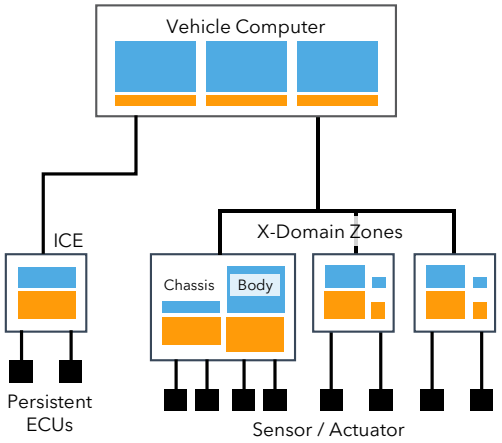
Domain + Body Zonal



Domain Controller / Vehicle Computer
- Higher level of control
- New features in software

Body Zonalization
- Wiring harness reduction, modularity introduced
- Primary / secondary body controllers

Cross-Domain Zonal

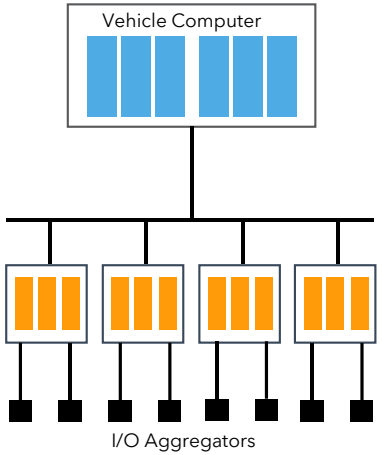


Vehicle Computer
- Consolidation, ECU reduction
- New features in software

X-Domain Zonalization
- Multi-tenancy, more ECU consolidation
- Compute local to zone

Persistent ECUs
- Limited value to migrate (e.g. too complex, lifespan)

Consolidated Compute



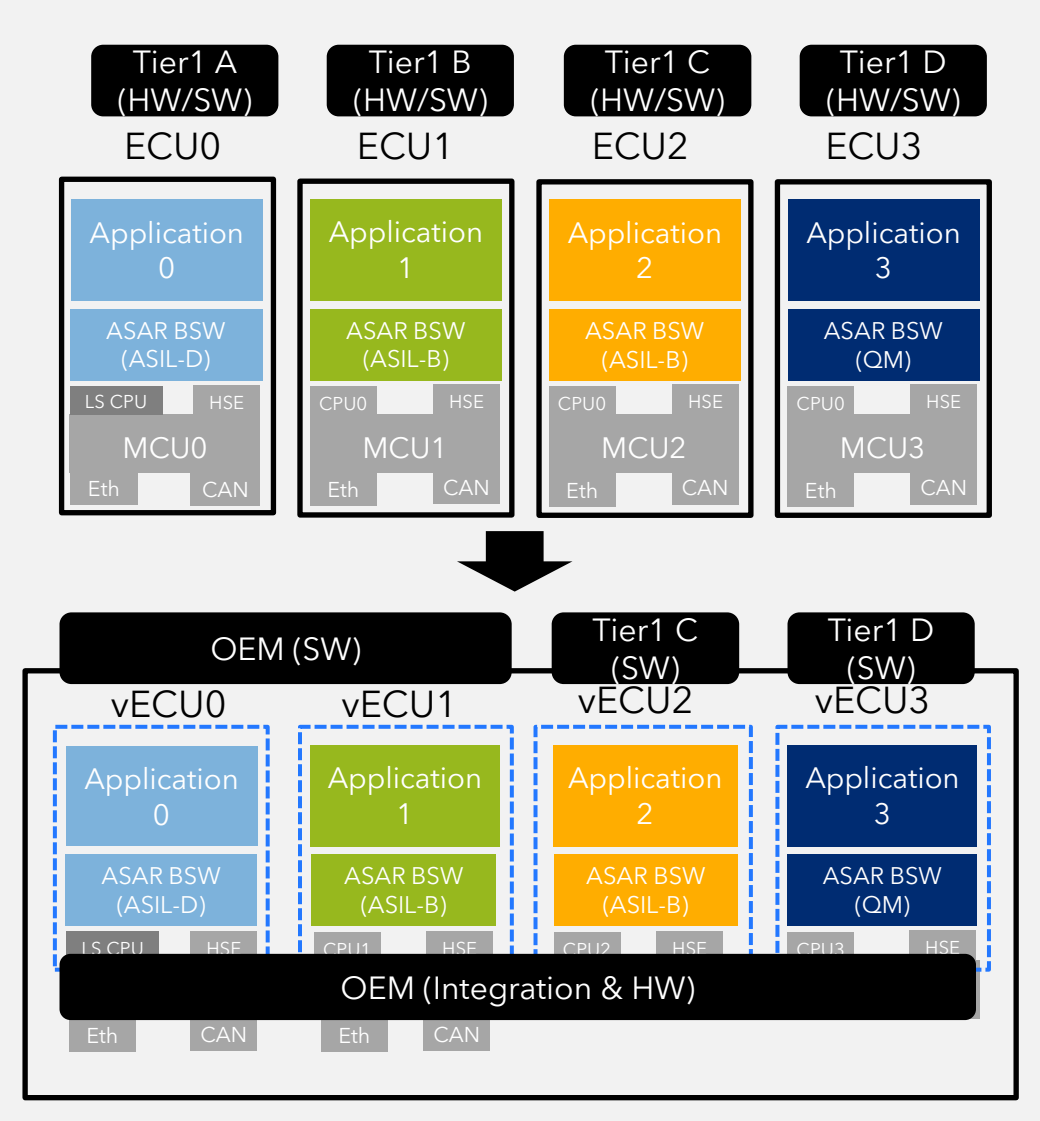
Vehicle Computer
- Hosts "All decision making"
- Fully flexible platform for SDV

I/O Aggregators
- Generic sensor / actuator control
- Available to multi-applications

Architectures focused on vehicle infrastructure – exclude ADAS and Infotainment

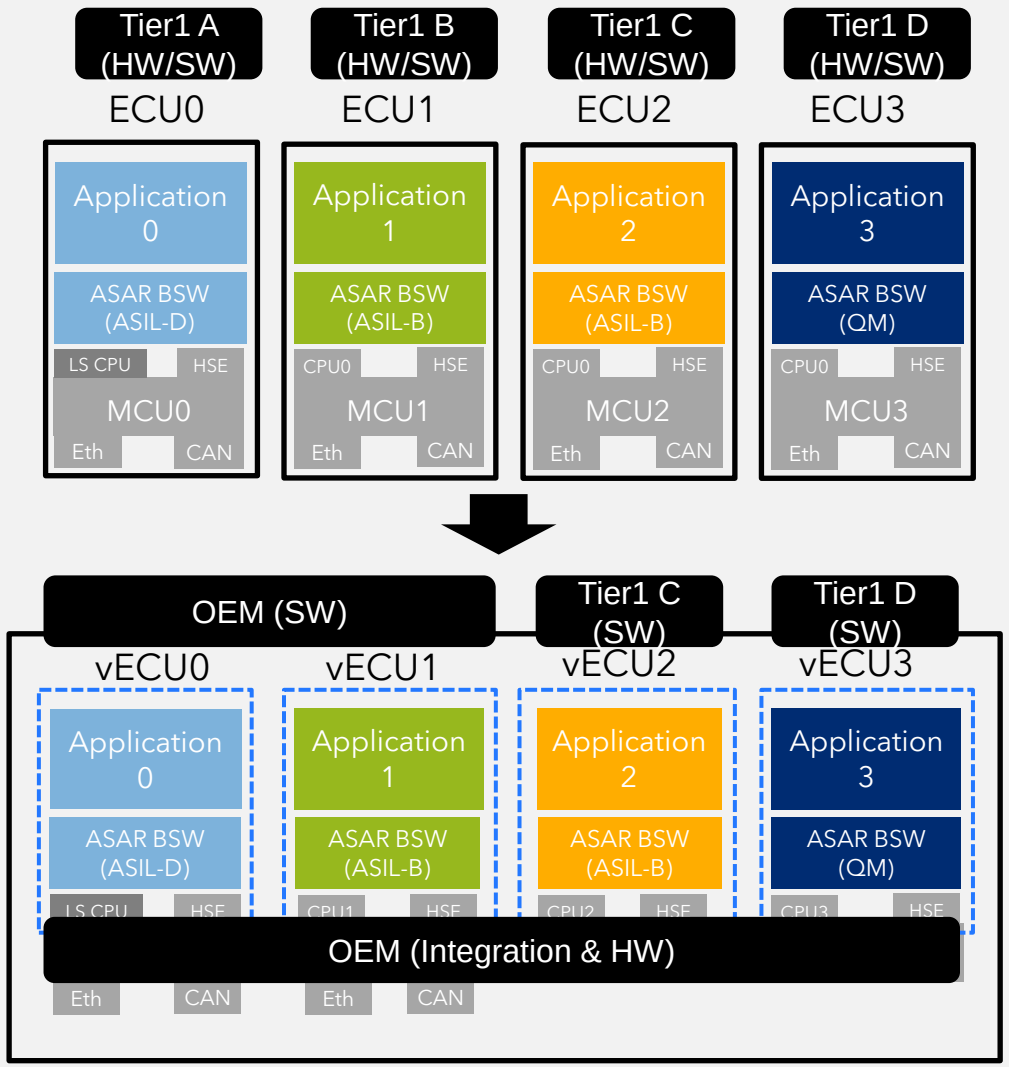
Embrace Virtualization

APPLICATION OWNERSHIP



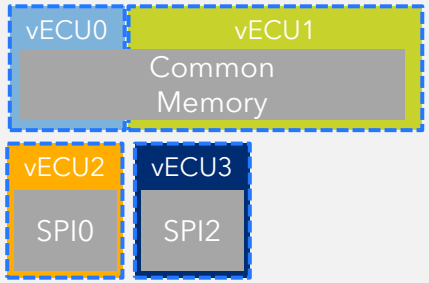
- New E/E architecture approaches aim to consolidate independent applications into a single SoC
- But software desires:
  - Virtual ECU = Physical ECU
  - Each SW environment believes it ‘owns’ the hardware
- Ownership of existing software a critical topic for ECU consolidation
- Evolving ecosystem between OEM, SW integrators and tier-one suppliers
- Virtualization on a single SoC provides protected, dedicated sandboxes for each vendor to contribute software

MIGRATION TO CONSOLIDATED ECU



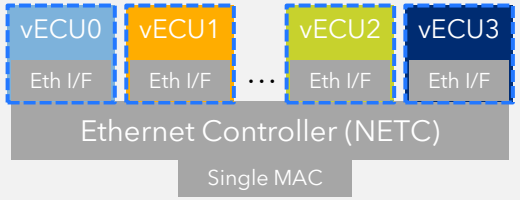
Full Chip Hardware Virtualization with Safe/Secure Resource Domain Partitioning

HW Access Control



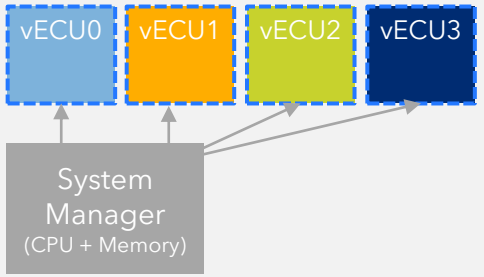
System-level firewall for all authorized accesses

HW Virtualized Peripherals



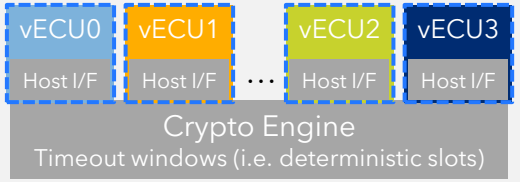
vECU software 'sees' dedicated resource, HW manages virtualization

System Manager



Partitioning and management of vECUs

HW Virtualized QoS



QoS mechanisms for shared resources

Software and Partner

NXP SOFTWARE OFFERING

- **VISION**: Accelerate adoption of transformational automotive architectures using S32x SoCs by providing leading class software tools, integrated frameworks and broad ecosystem of solutions within a common software architecture platform

ENABLING APPLICATIONS & ARCHITECTURES

Application and Reference Middleware
- Integrated PMIC/analog drivers
- Communications stacks: TCP/IP, AVB/TSN

Reference SW for future vehicle architectures
- Vehicle integration platform for cloud platforms
- FOTA, data analytics
- Domain and zonal middleware framework

EASE OF USE

Complete & robust enablement speeds development
- Real Time Drivers (RTD): AUTOSAR & non-AUTOSAR
- Security firmware standard
- Complex drivers for IP accelerators
- Support for RTOS and application OS's across Arm cores

NXP Software Ecosystem

SERVICES / APPLICATION SOFTWARE

MIDDLEWARE

OS / DRIVERS / SAFETY

HYPERVISOR (IF AVAILABLE)

ARM® CORTEX® CORES(S)

FIRMWARE / HW ACCELERATORS

NXP TOOLS & IDE

ACCELERATING CUSTOMER TIME TO MARKET

Full suite of tools for project integration:
- S32DS IDE & Config tool
- Premium Safety Software Framework (SAF)
- eIQ Auto AI/ML inference engine

S32x Platform Compatibility

NXP S32K | NXP S32Z | NXP S32G | NXP S32 Next

BROAD PARTNER NETWORK WITH STRONG AUTOMOTIVE EXPERTISE

The **Automotive Processing Partner Program** comprises a global network of independent companies that offer the vital tools, software, technology, engineering services and training to accelerate customer designs based on NXP components

Engineering ecosystem:

software and hardware engineering services, consultancy and training on products, safety and security and more.

Engineering Partners

Application ecosystem:

application software and networking stacks.

Application Partners

Enablement ecosystem:

OS & hypervisors, development tools, virtualization and modelling tools.

Enablement Partners

Rev. Sept 2023

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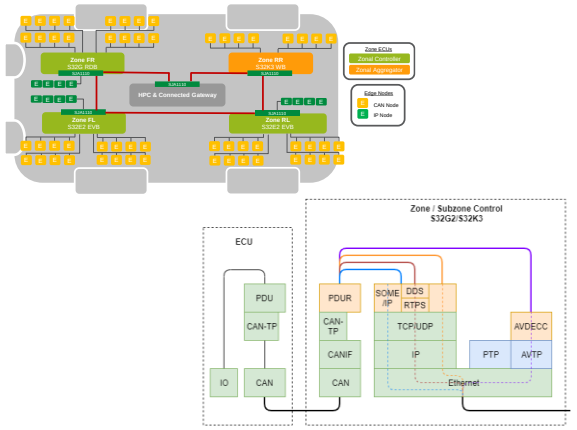
NXP DELIVERS APPLICATION EXPERTISE IN E/E ARCHITECTURES THROUGH MULTIPLE DEVELOPMENT AND DEMONSTRATION PLATFORMS

VIP reference designs

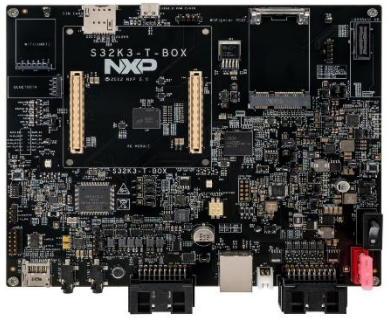


Zone POC platform

TSN Standards		
	IEEE 802.1Qbu	Frame Preemption
	IEEE 802.1Qbv	Scheduled Traffic
	IEEE 802.1Qci	Ingress Policing
	IEEE 802.1CB	Seamless Redundancy
	IEEE 802.1AS-REV	Time Synchronization
	IEEE 802.1Qcc	Stream Reservation

Ethernet TSN

Digital twin (BMS) demo



T-Box reference design

Smart data access demo

Motor control over network

Virtual development

BUILDING AN SDV

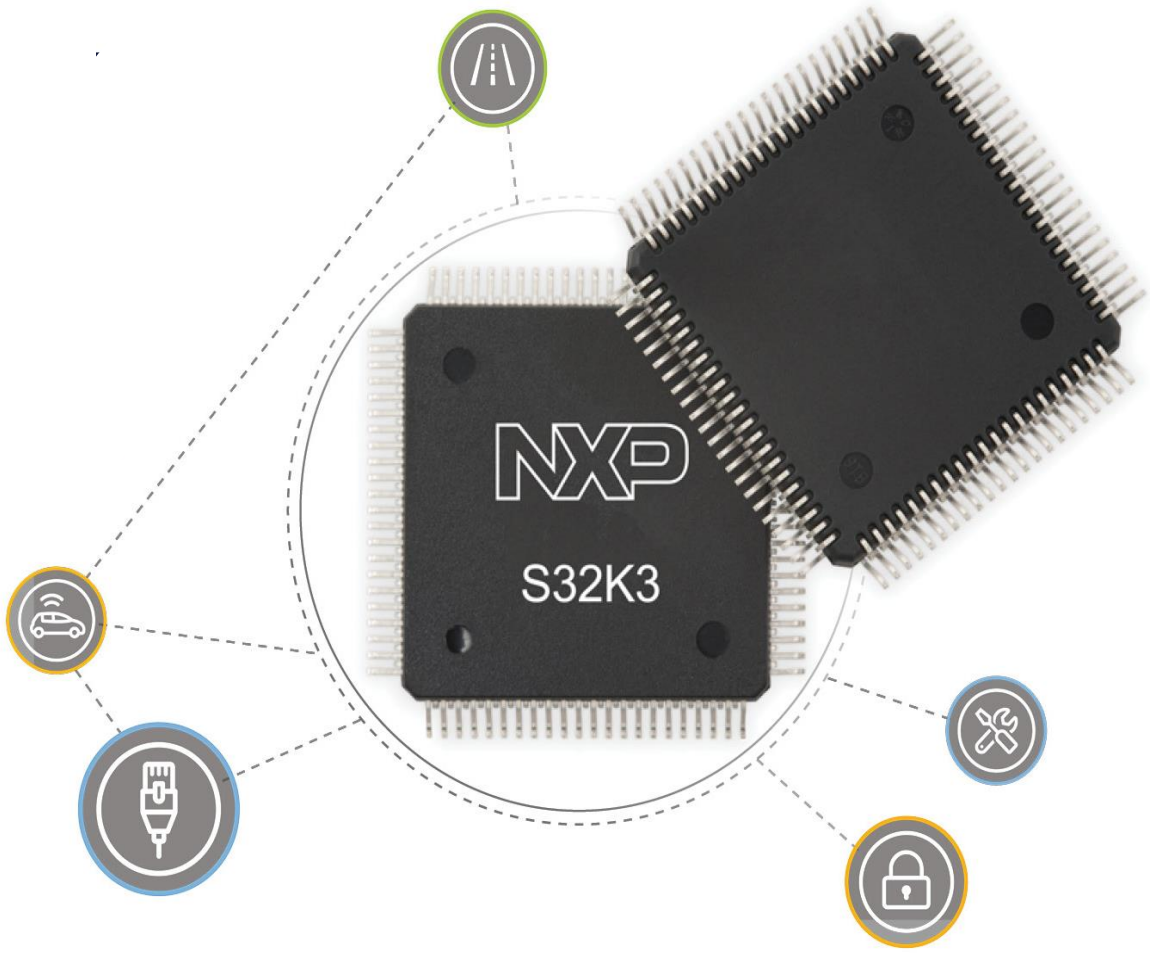
1. Select optimal E/E architecture
2. Re-architect vehicle software
3. Determine scope of vehicle platform
4. Adopt new Ethernet TSN technology
5. Embrace virtualization
6. Build in future-proofing
7. Architect vehicle-to-cloud software
8. Use virtual modelling
9. Select the right ecosystem partner(s)

S32K Microcontrollers

S32K3 AUTOMOTIVE MCU

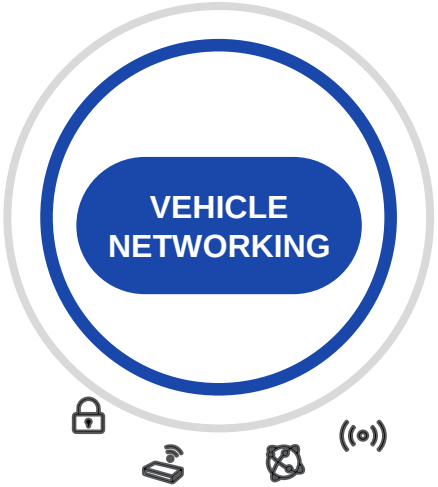
TACKLING SOFTWARE COST AND COMPLEXITY

- Security firmware and software streamlines development
- Safety hardware and software simplify ASIL certification
- Free AUTOSAR® drivers accelerate development and quality compliance
- Smart memory design enables fast and reliable over-the-air updates
- Breakthrough HDQFP package minimizes footprint



NXP S32 AUTOMOTIVE PROCESSING PORTFOLIO

● Production
● Development



Safety Everywhere Increasing Security

#1 General Purpose 32-bit
#1 Integrated Solutions
Leader in MC, body, BMS

● S32K1, S32K3 ●
● MagniV® → Next Gen. ●

xEV ,ENERGY Management Chassis & safety

Leader in xEV control
Highest performance real-time
Domain BMS, DPC, Inverters

● MPC57xxx
● S32E/Z, S32x ●

Exponential Data Growth Secure OTA UPDATES

#1 in Vehicle Network Processors
High-throughput networking, network security, service gateways, OTA updates

● S32G2, S32G3 ●
● S32x

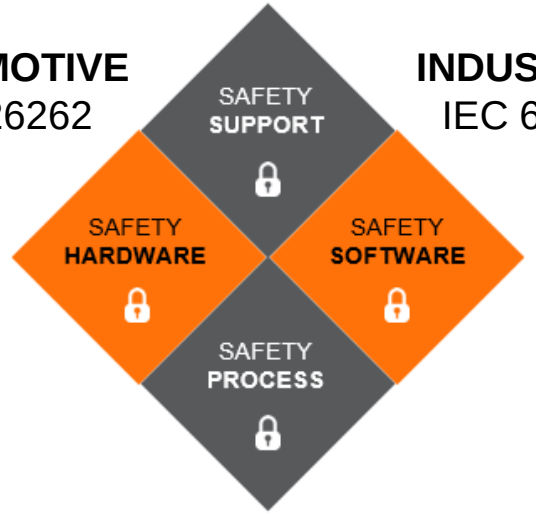
Scalable, Safe and Secure with Software, Tools and Compatible Analogue Products

FUNCTIONAL SAFETY, SECURITY AND LONGEVITY

SAFETY

AUTOMOTIVE
ISO 26262

INDUSTRIAL
IEC 61508



NXP QUALITY FOUNDATION

- Compliance with ISO 26262 / IEC 61508
- Reduced development complexity
- Safety hardware, software & documentation
- SafeAssure on-line community & direct safety expert support

SECURITY

- Compliance with ISO 21434
- Hardware accelerators with free software driver and firmware
- Security documentation & support – see ‘Secure Files’ option in NXP.com account (DocStore replacement)

LONGEVITY

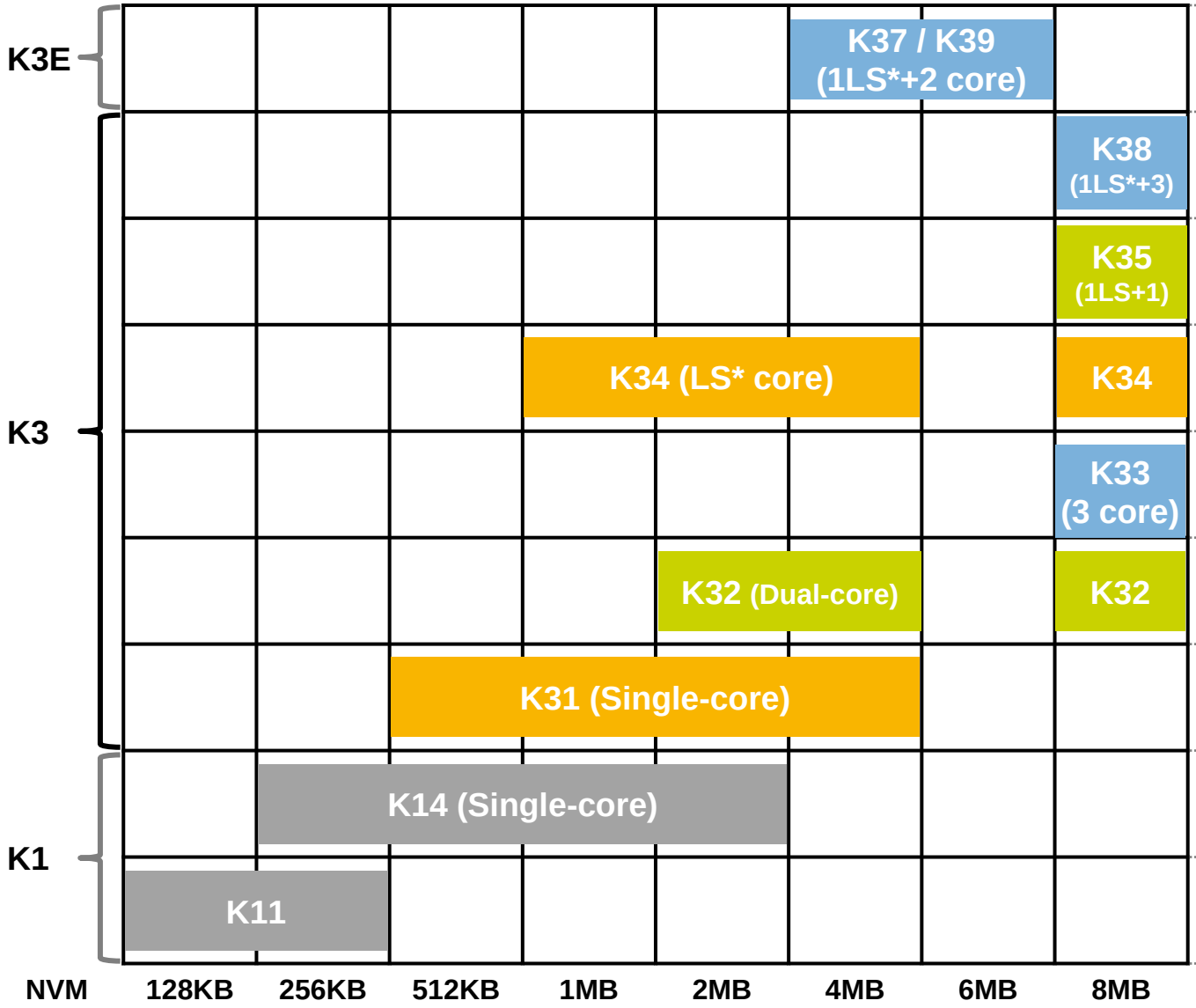
www.nxp.com/productlongevity

Product Longevity

Products	Launch Date	Longevity Years 10 Years	15 Years	Extended	Remains in Longevity Program until
S32K1					
S32K11x	Nov 2018	-	✓	✓	Nov 2036
S32K14x	Mar 2017	-	✓	✓	Mar 2036

- Minimum 15 years supply for all Automotive processors
- Lifecycle starts from product launch
- Strong historical commitment with recent extensions for key legacy processors

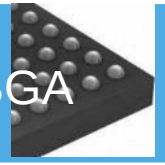
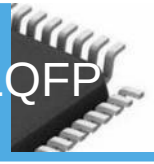
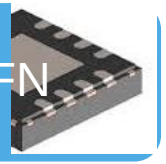
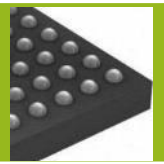
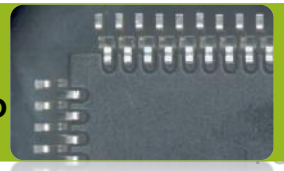
GPIS: BROAD SCALABLE PORTFOLIO AND APPLICATION RANGE



	Typical Application		Arm CPU Core	
ASIL D	Powertrain, OBC, Inverter, BMS	4 x M7 (1 x LS + 2) (2 x LS)	320MHz
ASIL D	Zonal Aggregator	5 x M7 (1 x LS + 3) (2 x LS + 1)	320MHz
ASIL D	BMS, Electrification	3 x M7 (1 x LS + 1)	240MHz
ASIL D	BMS, Electrification	2x M7 (1 x LS)	160MHz 240MHz
ASIL B	Body, Electrification	3 x M7	240MHz
ASIL B	High perf body, Zone & IO, Info/Audio	2 x M7	160MHz 240MHz
ASIL B	“K1 extension”, Security 2022+, OTA	1 x M7	120MHz 160MHz
ASIL B	CAN FD and Ethernet nodes	1 x M4	112MHz
ASIL B	Small CAN FD and LIN nodes	1 x M0+	48MHz

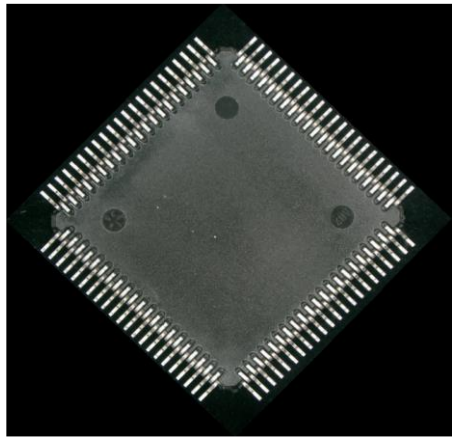
*S32K388 SPEC IS PRELIMINARY AND SUBJECT TO CHANGE, LS = lock step

S32K1 VS. S32K3

	S32K1	S32K3
Performance	M0+ @ 48MHz Or M4F @ 80-112MHz Single-core	1-5 M7 @ 120-320 MHz Single-core, Multi-core or Lockstep core
Memory	128KB-2MB P-Flash 17-256KB RAM	512KB-8MB P-Flash 128KB-1152KB RAM
Security & OTA	CSEc Sym Cipher Up to 20 keys OTA Support (RWW)	HSE B Sym & Asym Ciphers; 100+ Keys, Side channel protection Seamless OTA (RWW, Memory remapping for A/B Swap, FW rollback option)
Safety	ASIL B	ASIL B / D
Key Peripherals	100MbEthernet (AVB), up to 3 CAN FD Flextimer,TRGMUX,PDB for Motor control. FlexIO	Up to 1Gbps Ethernet (TSN, AVB), up to 8 CAN FD eMIOS, BCTU, LCU for Motor control Advanced peripherals I3C, Enhanced FlexIO
Packaging	BGA  LQFP  QFN 	BGA  HD QFP  LQFP

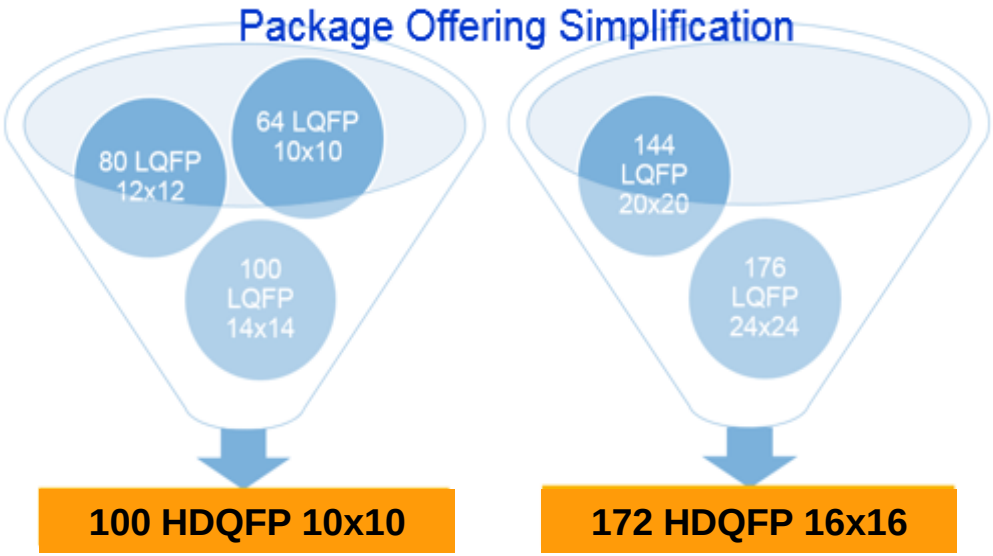
S32K3 PLATFORM – UNMATCHED SCALABILITY

- Core and platform: same Arm® Cortex®-M7 core across S32K3 family for SW reuse
- Security: HSE B across S32K3 family
- Safety: ASIL D and ASIL B fully compatible in S32K3 family
- Memory: 512K-8MB in S32K3, scalable down to 128KB in S32K1
- Package: BGA / HDQFP pin compatible in S32K3 family



K3 Flash	K3 Package				
	48 LQFP	100 HDQFP	172 HDQFP	257 BGA	289 BGA
	7 x 7 mm	10 x 10 mm	16 x 16 mm	14 x 14 mm	14 x 14 mm
8M			K358/48/38/28 *with Exposed Pad*		K388 K358/48/38/28**
6M					K376/K396**
4M			K344/24/14	K344/24/14	K374/K394**
2M		K342/22/12	K342/22/12		
1M	K311	K311	K341		
512K	K310	K310			

* Room temperature CES only, not for production
**BGA257 is subset of BGA289



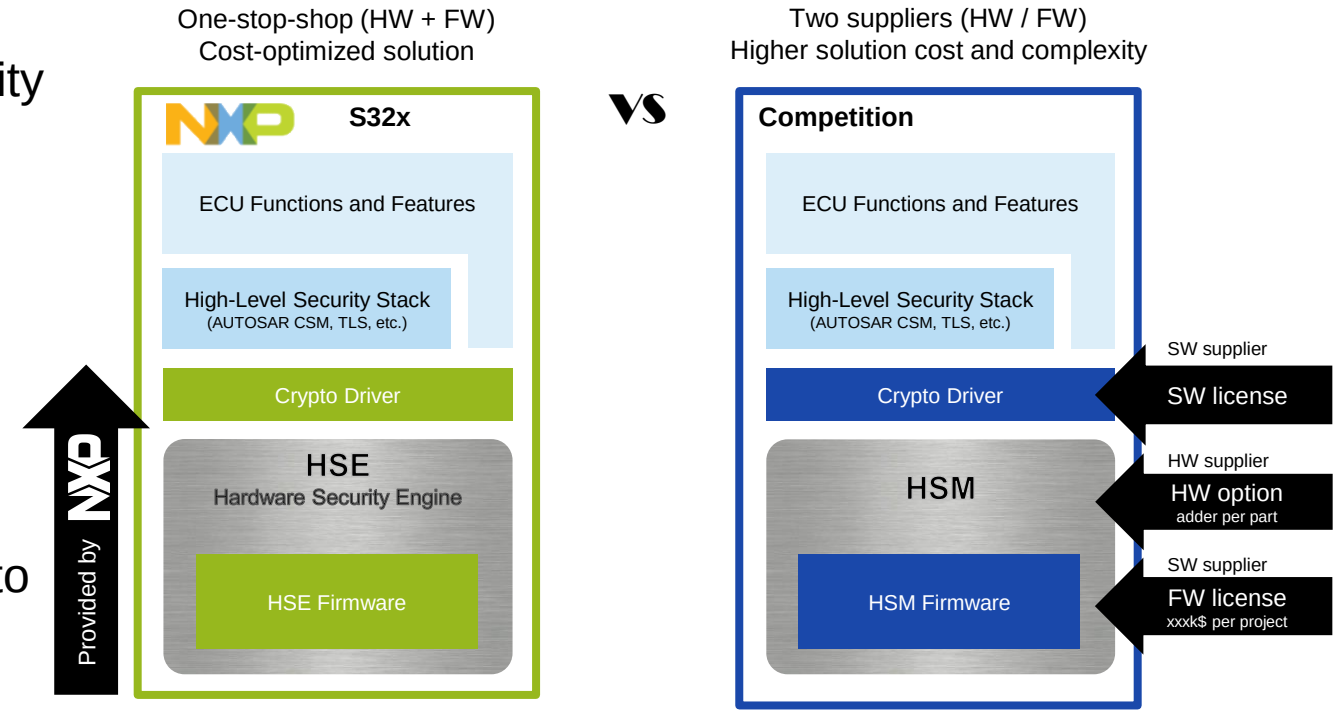
S32K3 SECURITY SOLUTION OVERVIEW

Description & Features:

- Firmware for dedicated on chip Hardware Security Engine (HSE) subsystem
- Provides extensive security services to host by offloading application processing
- Fully compatible with RTD crypto driver

Benefits:

- One stop shop solution: HSE + HSE FW + Crypto driver
- No license fees for standard HSE FW
- Covers all OEM security requirements
- Comprehensive services:
  - FAE support
  - PSIRT
  - FQE analysis



How to get it?

- Available on NXP website
- Refer business model & support section for more details

REAL-TIME DRIVERS

Description and features:

- Drivers for S32 MCU peripherals
- Two sets of APIs:
  - High Level Interface: based on MCAL. For AUTOSAR and Non-AUTOSAR Implementation
  - Low Level Interface: low level drivers. Contains IP-specific APIs. For Non-AUTOSAR implementation
- Supports AUTOSAR 4.4 and R21-11

Benefits:

- Production grade software
  - Production grade software with ASPICE and ISO-26262 compliance (up to ASIL D)
- Reduced development efforts
  - Single product covering AUTOSAR and Non-AUTOSAR environment
  - Scalable SW across S32 platform
  - Compatible with drivers of additional NXP ICs like PMIC, transceivers, etc.
- No additional charge
  - Supports multiple compilers: GCC, GHS, IAR, DIAB

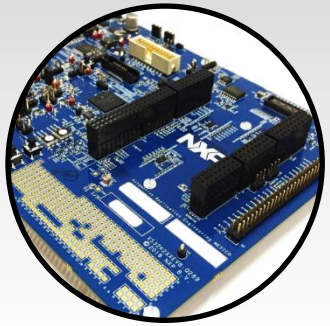
Real-Time Drivers

How to get it?

- Available on NXP website. www.nxp.com/rtd
- Refer business model and support section for more details

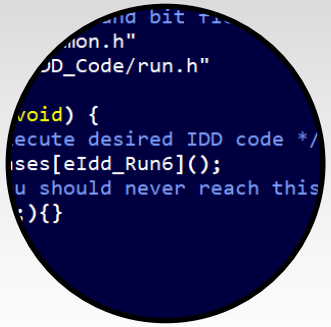
S32K3 SOLUTION – MAKE DEVELOPMENT AND PRODUCTION EASIER

Hardware Platform



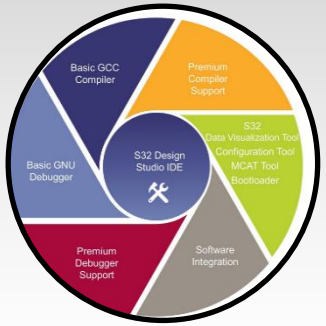
- EVB enables access to MCU full feature, basic debug with on board OpenSDA (no extra HW debugger needed)
- System reference design board(LED, Telematics)
- MCU + SBC solution level HW design guideline

Runtime Software



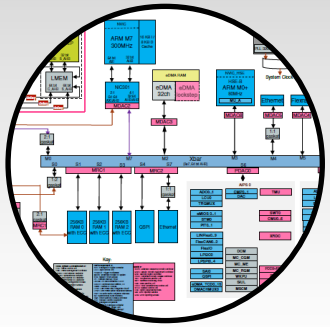
- Real-time driver (no extra cost)
- Security & OTA SW
- Safety framework + SCST SW
- Multi-core management SW
- Driver for SBC, BMS IC etc

Software Dev Tools



- S32DS IDE and config tool
- FreeMASTER, Model-based design
- Broad 3rd party support: IAR, GHS, Lauterbach, Isystem etc
- Power consumption management tool

Application Specific SW



- BMS
- Motor control
- Touch sensing
- LED lighting

Training and Support

TECHNOLOGY SHOWROOM

JOURNEYS BY DESIRED ENGAGEMENT

Self-guided tour
Live-streaming at set times
Guided tours

60+ VIRTUAL DEMOS

Focus on system solutions
Set up along NXP verticals

JOURNEYS BY DESIRED FOCUS

Edge & AI/ML
Safety & Security
Connectivity
Advanced Analog
Low Power Innovation
Sustainability

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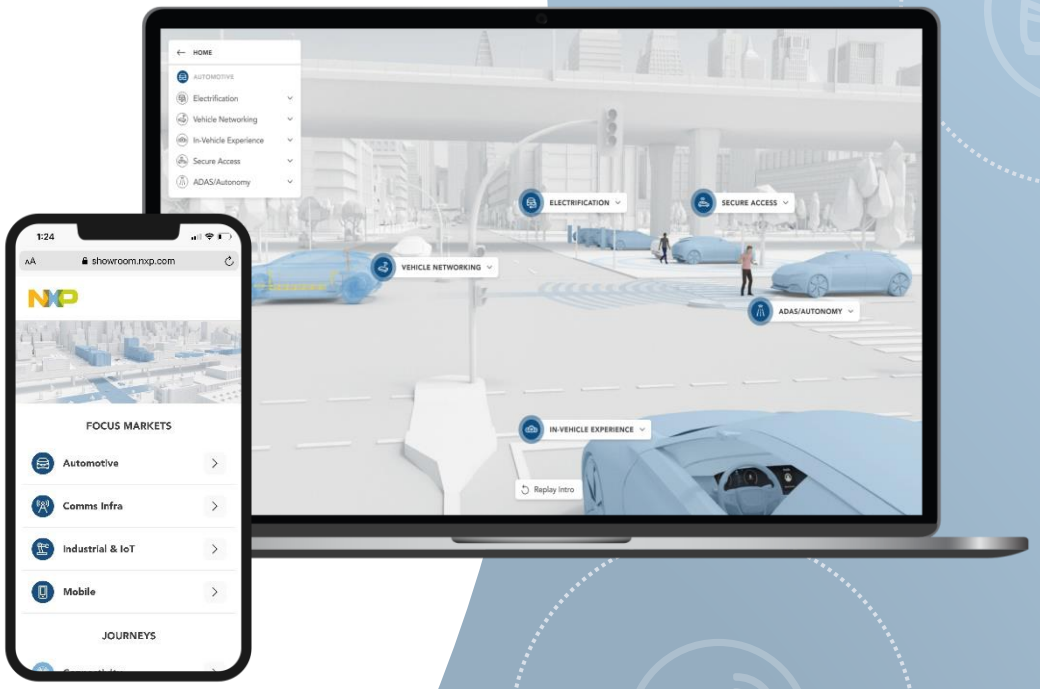

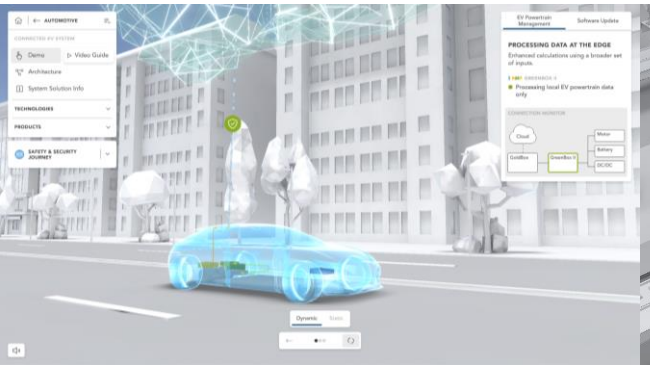

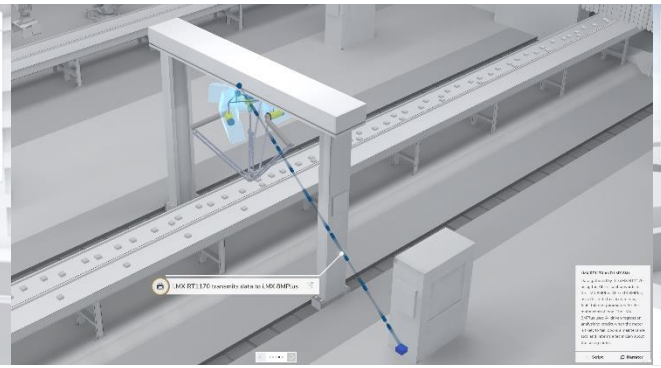

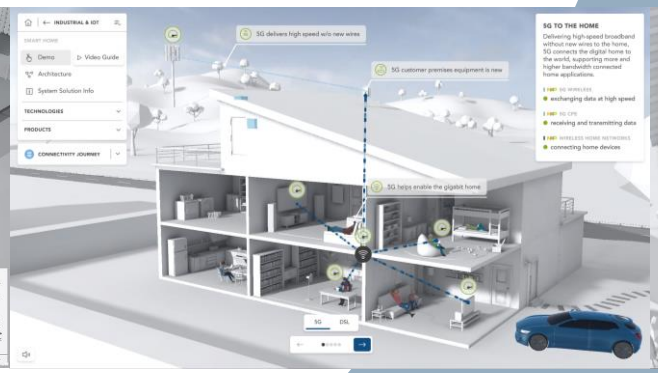

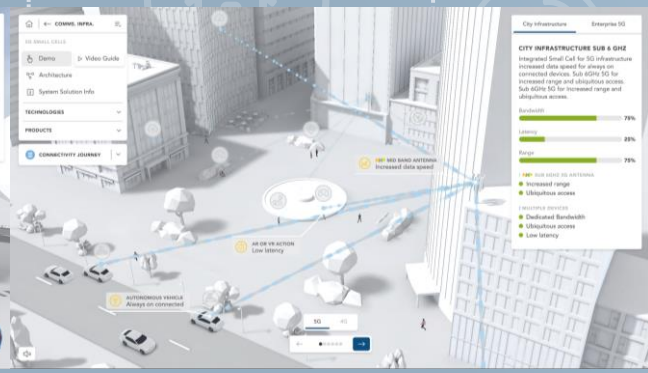

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